

Def: For a t.d.l.c.s.c group G , the (lower) rank-2 series, denoted by $(G_i)_{i \in \mathbb{N}}$, is defined as follows: $G_0 := G$ and $G_{i+1} := \text{Res}_2(G_i)$. We call a t.d.l.c.s.c. group rank-2 resolvable if the rank-2 stabilizes at $\{1\}$ in finitely many steps.

Thm: For G a t.d.l.c.s.c and $(G_i)_{i \in \mathbb{N}}$ the rank-2 series for G , the following hold:

(1) for each $i \geq 1$, $G_i = \text{Res}_{i+1}(G)$;

(2) $\text{Res}_\omega(G) = \bigcap_{i \in \mathbb{N}} G_i$;

(3) if G is compactly generated $\Rightarrow \text{Res}_\omega(G) = G_j$ for some $j \in \mathbb{N}$; in particular, the series $(G_i)_{i \in \mathbb{N}}$ stabilizes in finitely many steps.

$\hookrightarrow$ Proof:

(1) $\Rightarrow$ Let's argue by induction on $i \geq 1$. The case $i=1$ (base) holds by def. Suppose it works for i and consider G_{i+1} . Let $K \trianglelefteq G$ be s.t. $\varphi(G/K) \leq i+1$ and put $R := \pi^{-1}(\text{Res}_2(G/K))$ where $\pi: G \rightarrow G/K$ is the usual proj. We see that $(\text{Res}_2(G)) = G_i \leq R$ and that $\varphi(R/K) \leq i$ by some prop. The induction hypothesis implies that the i -th term of the rank-2-series for R

is contained in K . Since $0 < K$, we conclude that the i -th term of the rank-2 series of G is contained in K . That is to say, $0_{i+1} \leq K$, hence $0_{i+1} \leq \text{Res}_{i+1}(G)$.

The induction hyp. implies $\varphi(G_i/G_{i+1}) \leq i+1$, since 0_{i+1} is the i -th term of the rank-2 series for G_i . It is easy to see that $\text{Res}_i(G/G_{i+1}) = G_i/G_{i+1}$, hence applying some prop, $\varphi(G/G_{i+1}) \leq i+2$. Therefore $G_{i+1} \geq \text{Res}_{i+2}(G) \Rightarrow G_{i+1} = \text{Res}_{i+1}(G)$.

(2) $\Rightarrow$ G/G_j is the largest quot of G of rank at most $j+1$ by claim (1). In particular, if G/K is a quot of G of fin. rank $\Rightarrow K \geq G_j$ for some j . Hence $\bigcap_{j \in \mathbb{N}} G_j \leq \text{Res}_\omega(G)$. The other inclusion is obvious.

(3) $\Rightarrow$ One of the props implies $G/\text{Res}_\omega(G)$ has finite rank $\Rightarrow \text{Res}_\omega(G) = G_j$ for some j . ■

Coro: For G a t.d.l.c. n.c group, G is elementary group with finite rank iff G is rank-2 reducible.

Coro: Let G be an elementary group s.t. $\varphi(G) < \omega$, let $(G_i)_{i \in \mathbb{N}}$ be the rank-2 series for G , and let $i \in \mathbb{N}$. Then

$$\varphi(G) = (\varphi(G_i) + 1) + \varphi(G/G_i)$$

↳ Proof:

& $G_i = \text{Res}_{i+1}(G_i)$ and $\exists 0 \leq j \leq i$ s.t. $g(G/G_i) = j+1$. It follows that $G_i = \text{Res}_{i+1}(G) = \text{Res}_{j+1}(G) = G_j$ and that there is a strictly decreasing chain of ~~the~~ subgroups $G_0 > G_1 > \dots > G_j$. Repeated application^{one} them gives us that $g(G) = g(G_j) + j$. We thus deduce that $g(G) = g(G_j) + j = (g(G_j) - 1) + g(G/G_j)$ verifying the corollary. ■

Let T be a countably regular tree and fix δ the end of T . We orient the edges of T s.t. all the edges point towards the end δ . The resulting directed graph is denoted $\vec{T}$; we call δ the distinguished end of $\vec{T}$. Given a countable set X , a coloring of $\vec{T}$ is a function $c: E\vec{T} \rightarrow X$ s.t. for each $v \in V\vec{T}$,

$$c_v := c|_{\text{inn}(v)}: \text{inn}(v) \rightarrow X$$

is a bijection. The set $\text{inn}(v)$ is the collection of directed edges with terminal vertex v . The coloring allows us to define the local action of $g \in \text{Aut}(\vec{T})$ at $v \in V\vec{T}$:

$$\sigma(g, v) = c_{g(v)} \circ g \circ c_v^{-1} \in \text{Sym}(X)$$

(?) We call the coloring ended if there is a monochromatic directed ray which is a representative of the distinguished end δ ; we always assume the coloring is ended.

The tree $\vec{T}$ is equipped with horoballs and horospheres centered at δ , defined as follows. There is a function $\eta: V\vec{T} \rightarrow \mathbb{Z}$ s.t., for each of the oriented edges e of $\vec{T}$, we have $\eta(t(e)) = \eta(o(e)) + 1$. One can see that η is unique up to an additive constant; it is determined by choosing a basepoint v_0 s.t. $\eta(v_0) = 0$. The set $\{\eta^{-1}(i) | i \in \mathbb{Z}\}$ of fibers is therefore uniquely determined.

The fibers are the horospheres centered at δ , and the sets $X_i := \{v \in V\vec{T} : \eta(v) > i\}$ for $i \in \mathbb{Z}$ are the horoballs centered at δ .

Def: (t.d.l.c. s.c. permutation grp)

A pair (G, X) where G is a t.d.l.c. s.c. group, $|X| = \aleph_0$ and $G \curvearrowright X$ faithfully with compact-open ^{point} stabilizers.

Def: $(E_X(G, U))$

Suppose that (G, X) is a t.d.l.c. s.c. permutation grp with $U \in \mathcal{U}(G)$ and color the tree $\vec{T}$ by X . We define the group $E_X(G, U) \leq \text{Aut}(\vec{T})$ as follows: $E_X(G, U)$ is the set of $g \in \text{Aut}(\vec{T})$ s.t. $\sigma(g, v) \in G$ for all $v \in V\vec{T}$ and that $\sigma(g, v) \in U$ for all but finitely many $v \in V\vec{T}$. $\rightarrow$ 'are you changing colors?' $\rightarrow$ 'Aut is as directed?'

It is easy to verify that $E_X(G, U)$ is an abstract grp.

For a vertex $v \in V\vec{T}$, it is an easy exercise to check that the pt stabilizer $E_X(G, U)_{(v)}$ is compact in the topology inherited from $\text{Aut}(\vec{T})$.

$\rightarrow \{(g_i)_{i \in \mathbb{N}} \subseteq E_X(G, U)_{(v)}, \text{ let } w \in \vec{T}$

$\sigma(g_i, w)$

Think

(The top on $\text{Aut}(\vec{T})$ is given by pointwise convergence: it is Polish but not loc. compact).